

Introduction to JS

JS is a high level programming language, mainly used for web dev. It does interactive web algo. applications, supporting both client & server side develop & interacting seamlessly with HTML, CSS and a rich standard library.

JS can interact with DOM - (Document object model)

to modify the structure, styling, content of the webpages. It is often used with Node.js, a runtime environment which allows JS codes to be executed at the outside of the browser.

- ↳ single threaded lang that executes one task at a time.
- ↳ an interpreted language that executes coded line by line.
- ↳ data type of the variable is decided at run-time, that's why its called dynamically typed.

Key features of JavaScript

- (i) Client-side scripting - JS runs on the user's browsers, so has a faster response time without communicating the server.

(ii) Event driven - It can respond to user actions. { clicks, keystrokes } in real time.

(iii) Asynchronous - JS can handle tasks like fetching data from servers.

↳ Client-side nature of JS -

- Involves controlling the browser & its DOM
- Handles user events like clicks & form inputs
- Common library includes Angular JS, React JS & vite JS.

↳ Server-side nature of JS -

- Involves interacting with databases, manipulating files & generating responses.
- Node JS & other frameworks like Express JS are widely used for server-side JS enabling full stack development.

↳ Application of JS -

JS is used in a wide range of applications, from enhancing websites to building complex applications, like -

(i) Web development - JS adds interactivity & dynamic behaviour to static websites with popular frameworks like angular JS, React JS etc enhancing development.

(ii) Server Application - Node JS brings JS to the server side, enabling powerful server application & full stack develop

(iii) Game development - JS combines with HTML 5 (five) & libraries like Ease JS enables creation of interactive games for the web.

~~X~~ External JavaScript - this code can be placed at a diff. file, means external (.js) file linked with an HTML document.

for Example

Folder Name $\Rightarrow$ My First HTML

$\begin{cases} \rightarrow \text{Index.html} \\ \rightarrow \text{Index.js} \end{cases}$

Inside Index.html -

```
<!DOCTYPE html>
```

```
<html lang = "en">
```

```
<head>
```

```
<script src = "Index.js"> </script>
```

```
</head>
```

```
<body>
```

```
<p> My first HTML code </p>
```

```
<p> Namaste Dunia! </p>
```

```
</body>
```

```
</html>
```

Inside Index.js -

```
function my-first-function () {
```

```
    alert ("Hello!");
```

```
}
```